Formal Uncertainty Propagation for Stochastic Dynamical Systems with Additive Noise

Steven Adams, Eduardo Figueiredo and Luca Laurenti

Delft University of Technology

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Uncertainty Propagation for Stochastic Dynamical Systems

Stochastic Dynamical Model

$$x_{k+1} = f(x_k) + \omega_k, \qquad x_0 \sim \mathbb{P}_{x_0}, \quad \omega_k \sim \mathbb{P}_\omega \tag{1}$$

Motivating Applications

- Planning and verification under uncertainty
- Stochastic and distributionally robust MPC

Practical Challenge

The initial and noise distributions are generally unknown and must be estimated (from data), introducing **distributional uncertainty**.

Modeling Uncertain Initial and Noise Distributions

ρ-Wasserstein Ambiguity Sets

$$\mathbb{B}_\theta(\mathbb{P}) = \{\mathbb{Q} \in \mathcal{P}_\rho(\mathcal{X}) \mid \mathbb{W}_\rho(\mathbb{P}, \mathbb{Q}) \leq \theta\} \subset \mathcal{P}_\rho(\mathcal{X}).$$

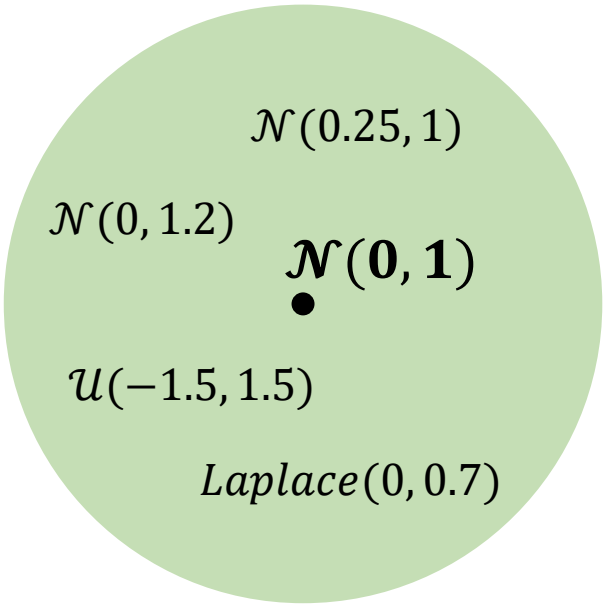


Figure: Illustration of distributions within radius 0.25 in 2-Wasserstein from $\mathcal{N}(0, 1)$.

Propagating Uncertainty from Ambiguous Distributions

Uncertain Stochastic Dynamical Model

For uncertainty radii $\theta_{x_0}, \theta_\omega$ and Gaussian mixture centers $\bar{\mathbb{P}}_{x_0}, \bar{\mathbb{P}}_\omega$:

$$x_{k+1} = f(x_k) + \omega_k, \quad x_0 \sim \mathbb{P}_{x_0} \in \mathbb{B}_{\theta_{x_0}}(\bar{\mathbb{P}}_{x_0}), \quad \omega_k \sim \mathbb{P}_\omega \in \mathbb{B}_{\theta_\omega}(\bar{\mathbb{P}}_\omega)$$

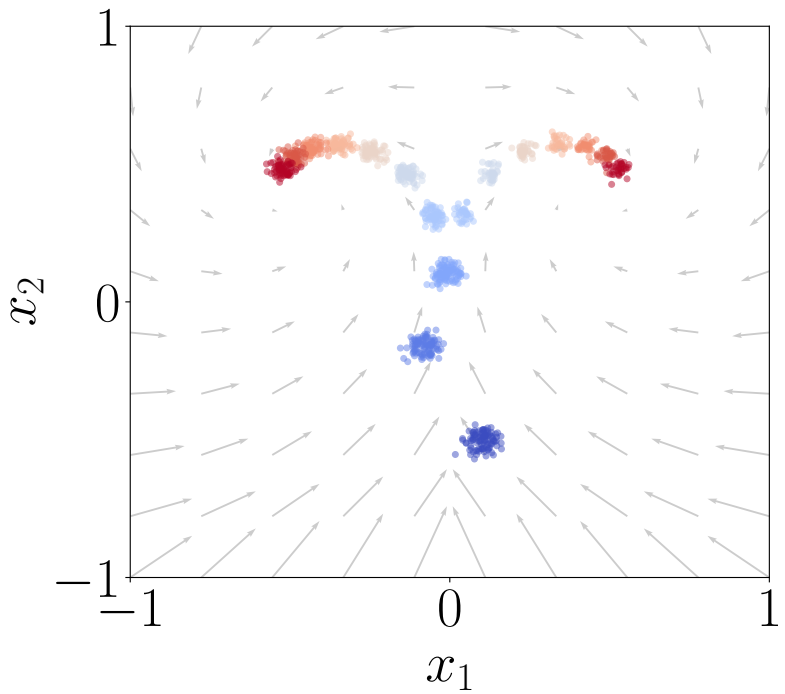


Figure: Dynamics f as vector fields. Samples from the true distributions evaluated over 20 time steps, with a blue-red gradient indicating time evolution.

Propagating Uncertainty from Ambiguous Distributions

Uncertainty Propagation Problem

For horizon K, compute sets $\mathbb{S}_{x_0}, \ldots, \mathbb{S}_{x_K}$ defined recursively for $k < K$:

$$\mathbb{S}_{x_0} = \mathbb{B}_{\theta_{x_0}}(\bar{\mathbb{P}}_{x_0}),$$
$$\mathbb{S}_{x_{k+1}} = \{(f\#\mathbb{P}_{x_k}) * \mathbb{P}_{\omega_k} \mid \mathbb{P}_{x_k} \in \mathbb{S}_{x_k}, \mathbb{P}_{\omega_k} \in \mathbb{B}_{\theta_\omega}(\bar{\mathbb{P}}_\omega)\}.$$

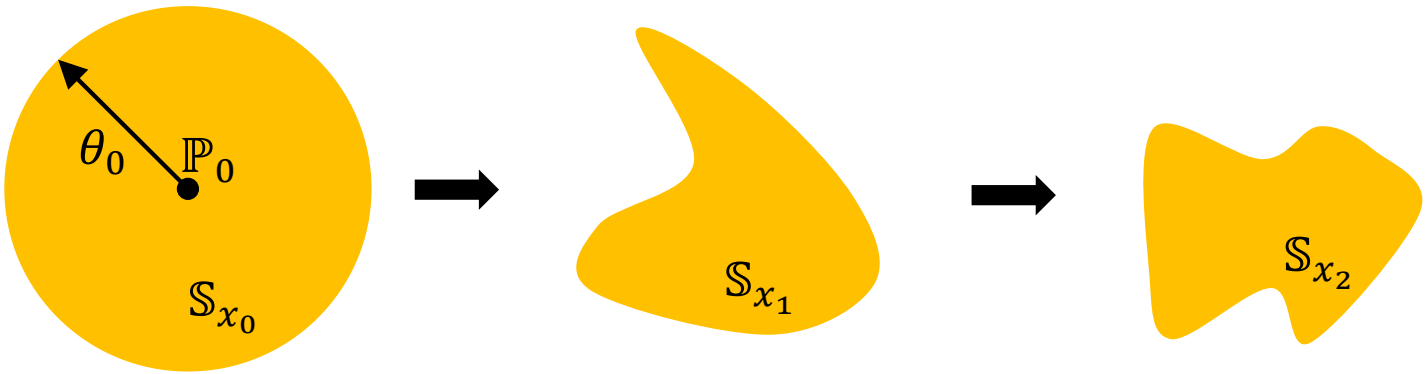
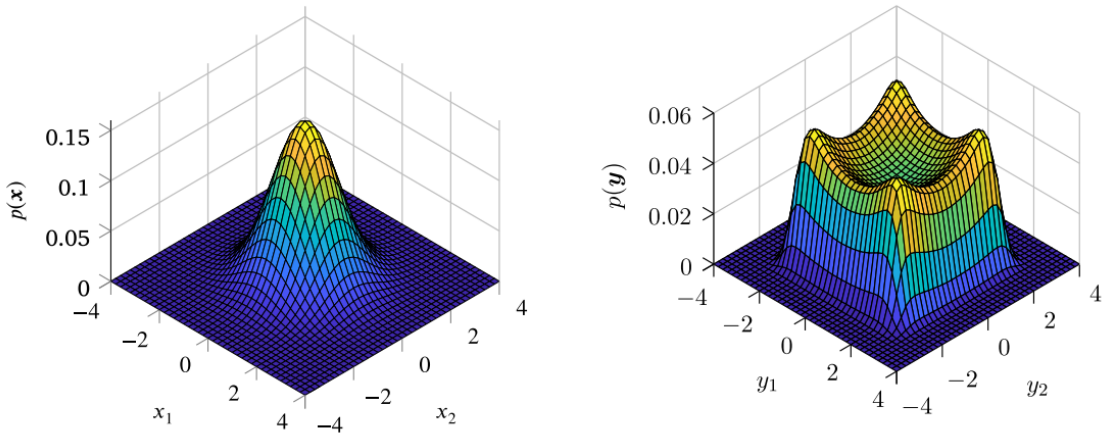


Figure: Schematic illustration of the uncertainty propagation problem.

Propagating Uncertainty from Ambiguous Distributions

Already for $\theta_{x_0}, \theta_\omega = 0$ (i.e. propagating probability distributions instead of ambiguity sets), (1) is generally intractable[1].



(a) Gaussian distributed random variable.

(b) Nonlinearly transformed random variable.

Figure: Example of non-linear propagation, with $f(x) = 2\tan(x) + Ax$, and $x \sim \mathcal{N}(0, I)$ (more details in [2])

[1] J. Quiñonero-Candela, A. Girard, and C. E. Rasmussen (2003). *Prediction at an Uncertain Input for Gaussian Processes and Relevance Vector Machines Application to Multiple-Step Ahead Time-Series Forecasting*. Tech. rep.

[2] D. Landgraf, A. Völz, F. Berkel, K. Schmidt, T. Specker, and K. Graichen (2023). "Probabilistic prediction methods for nonlinear systems with application to stochastic model predictive control". In: *Annual Reviews in Control* 56, p. 100905

Propagating Uncertainty from Ambiguous Distributions

Method	Guarantees	Scalable	Non-linear
Moment-matching[1]	-	✓	✓
Filtering [2]	-	~	✓
Numerical integration[3]	-	-	✓
Linear OT[4]	✓	✓	-
Total Variation[5]	✓	-	✓
Quantization-Based[6]	✓	~	✓

[1] M. Deisenroth and C. E. Rasmussen (2011). "PILCO: A model-based and data-efficient approach to policy search". In: *Proceedings of the 28th International Conference on machine learning (ICML-11)*, pp. 465–472

[2] S. J. Julier and J. K. Uhlmann (2004). "Unscented filtering and nonlinear estimation". In: *Proceedings of the IEEE* 92.3, pp. 401–422

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[5] E. Figueiredo, A. Patane, M. Lahijanian, and L. Laurenti (2024). "Uncertainty Propagation in Stochastic Systems via Mixture Models with Error Quantification". In: *2024 IEEE 63rd Conference on Decision and Control (CDC)*

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Our Approach: Quantization-Based Uncertainty Bounds

For each time-step k

1. Build center distribution $\bar{\mathbb{P}}_{k+1}$ via quantization of $\mathbb{P}_k$.
2. Compute radius θ_k ensuring $\mathbb{S}_{k+1} \subseteq \mathbb{B}_{\theta_{k+1}}(\bar{\mathbb{P}}_{k+1})$.

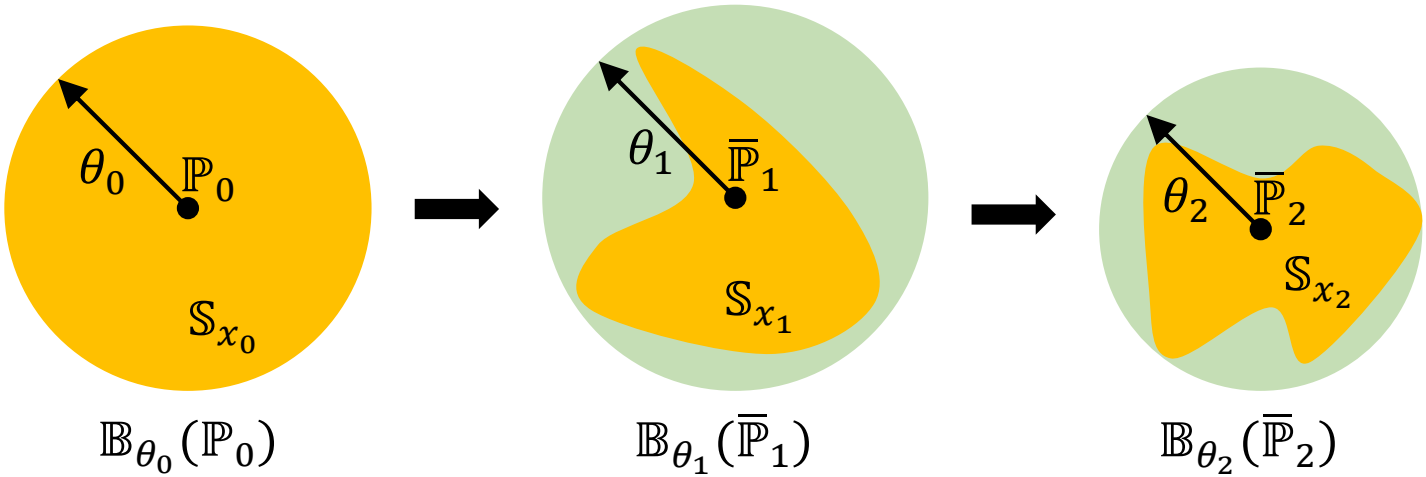


Figure: Schematic illustration of the approach.

Quantizing Continuous into Discrete Distributions

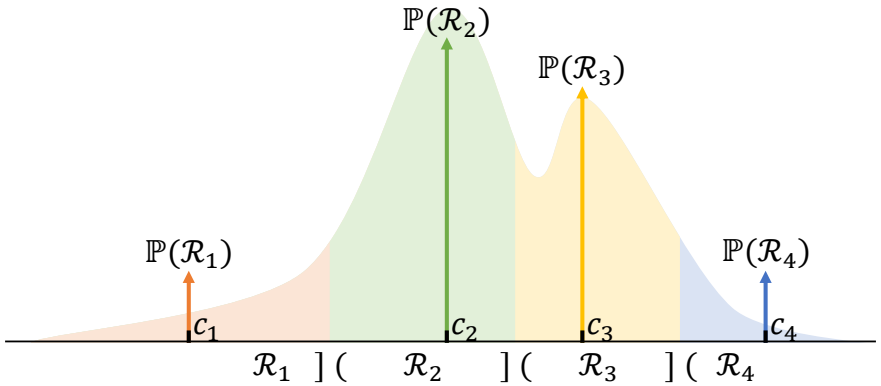


Figure: Quantization of a continuous distribution.

Quantization Operation

For a set of points $\mathcal{C} = \{c_i\}_{i=1}^N \subset \mathbb{R}^d$, define $\Delta_{\mathcal{C}} : \mathbb{R}^d \to \mathcal{C}$ by

$$\Delta_{\mathcal{C}}(z) = \arg\min_{c \in \mathcal{C}} \|z - c\|.$$

Quantizing Continuous into Discrete Distributions

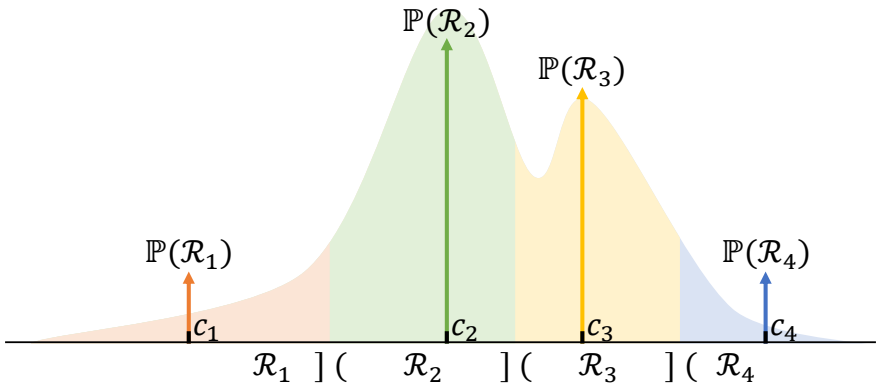


Figure: Quantization of a continuous distribution.

Quantization of a Probability Distribution

Let $\{\mathcal{R}_i\}_{i=1}^N$ be the Voronoi partition induced by $\mathcal{C}$. The quantization of $\mathbb{P} \in \mathcal{P}(\mathbb{R}^d)$ w.r.t. $\mathcal{C}$ is

$$\Delta_{\mathcal{C}} \# \mathbb{P} = \sum_{i=1}^{N} \mathbb{P}(\mathcal{R}_i)\, \delta_{c_i}.$$

Quantizing Continuous into Discrete Distributions

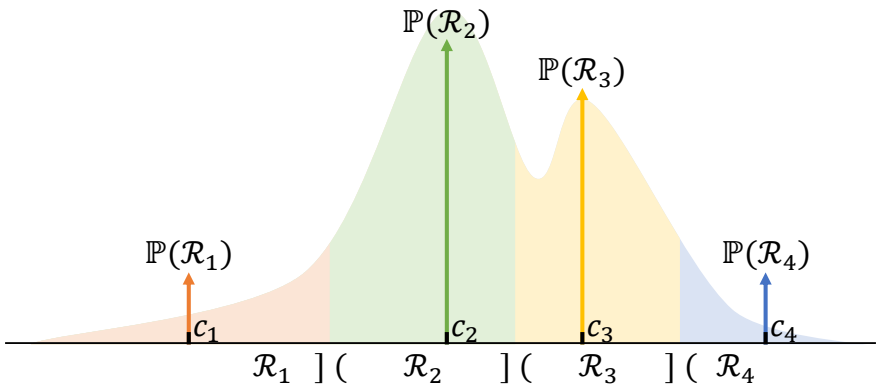


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Quantization Error

Under the optimal transport plan, mass in each region R_ℓ moves to its center c_ℓ, yielding

$$\mathbb{W}_\rho(\mathbb{P}, \Delta_{\mathcal{C}} \# \mathbb{P}) = \left(\sum_{\ell=1}^{|\mathcal{C}|} \int_{\mathcal{R}_\ell} \|x - c_\ell\|^\rho \, d\mathbb{P} \right)^{1/\rho}.$$

Ambiguity Ball Centers via Gaussian Mixtures

Using the quantization operation[7] defined at each time k by the locations $\mathcal{C}_k$, we define the center of the ambiguity set at time $k+1$ as

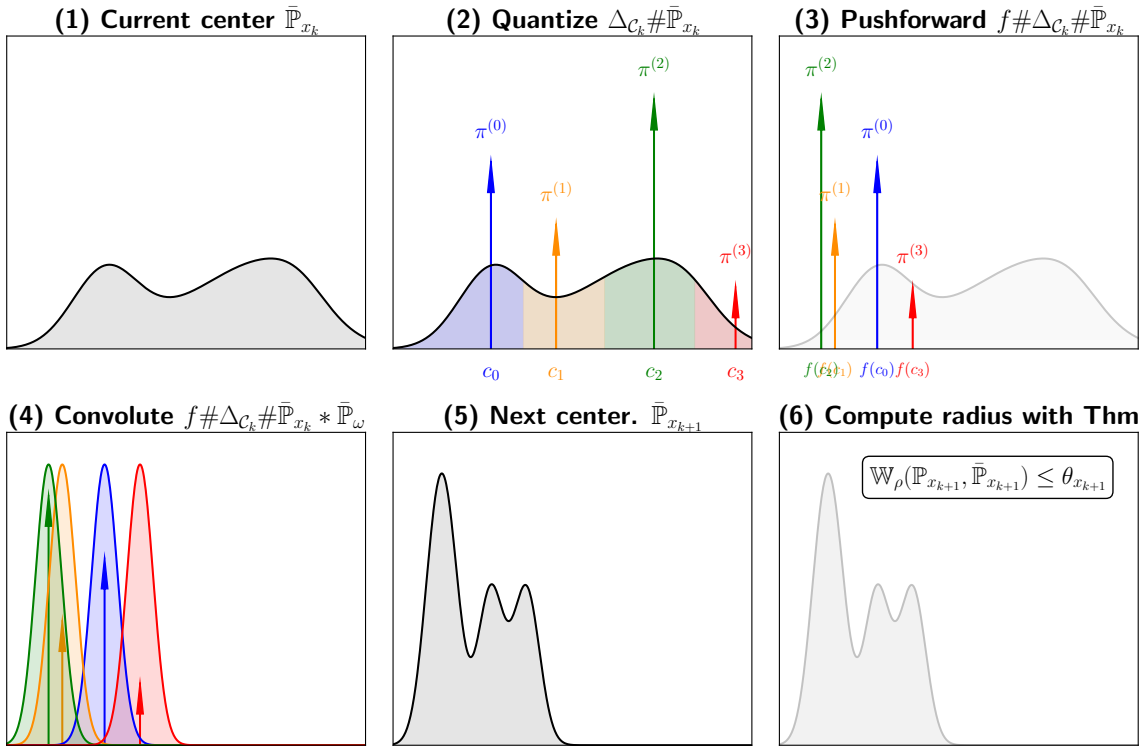


Figure: Ambiguity ball propagation.

[7] S. Adams, E. Alwash, and L. Laurenti (2025). "discretize_distributions: Efficient Quantization of Gaussian Mixtures with Guarantees in Wasserstein Distance". In: *arXiv preprint arXiv:2511.15854*

Ambiguity Radii via Local Bounds on Dynamics

Theorem Using a (Global) Lipschitz Constant

For each $k \in \{0, \ldots, K-1\}$, let

$$\theta_{\Delta,k} = \mathbb{W}_\rho(\bar{\mathbb{P}}_{x_k}, \Delta_{c_k} \# \bar{\mathbb{P}}_{x_k}). \tag{2}$$

Then define

$$\theta_{x_{k+1}} = \theta_\omega + \mathcal{L}_f(\theta_{x_k} + \theta_{\Delta,k}). \tag{3}$$

For all k, it holds that $\mathbb{S}_{x_k} \subseteq \mathbb{B}_{\theta_{x_k}}(\bar{\mathbb{P}}_{x_k})$.

Ambiguity Radii via Local Bounds on Dynamics

Consider locations $\mathcal{C}_k = \{c_{k,\ell}\}_{\ell=1}^{N_k}$. For each ℓ, choose coefficients $\alpha_{k,\ell}, \beta_{k,\ell} \in \mathbb{R}_{\geq 0}$ such that for all x

$$\|f(x) - f(c_{k,\ell})\|^\rho \leq \alpha_{k,\ell}\|x - c_{k,\ell}\|^\rho + \beta_{k,\ell}. \tag{2}$$

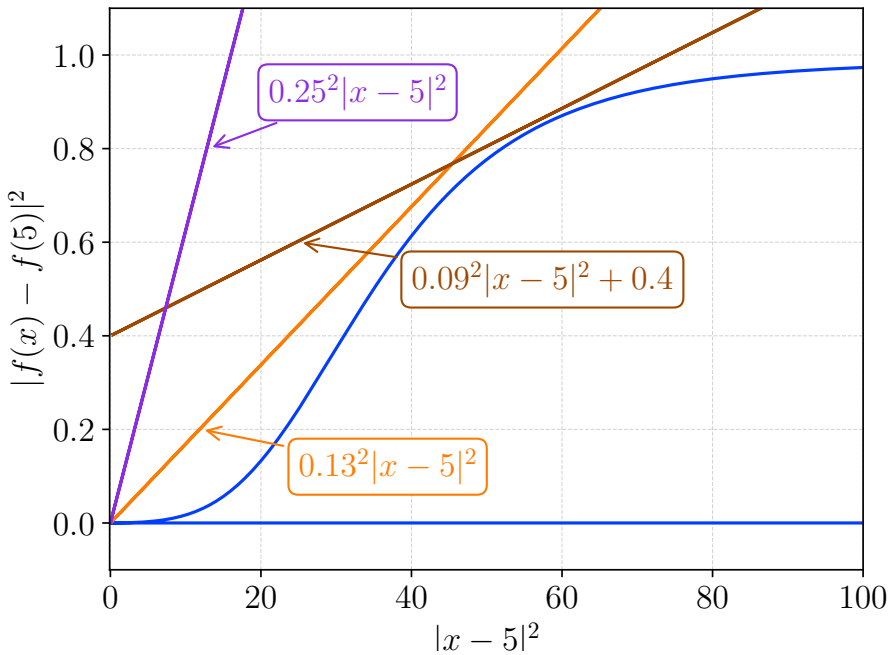


Figure: Examples of pairs (α, β) for sigmoidal dynamics f.

Ambiguity Radii via Local Bounds on Dynamics

Theorem

For each $k \in \{0, \ldots, K-1\}$, let $\hat{\alpha}_k = \max_\ell \alpha_{k,\ell}$. Then define

$$\theta_{x_{k+1}} = \theta_\omega + \left(\hat{\alpha}_k (\theta_{x_k} + \theta_{\Delta,k})^\rho + \sum_{\ell=1}^{N_k} \bar{\mathbb{P}}_{x_k}(\mathcal{R}_{k,\ell}) \beta_{k,\ell} \right)^{1/\rho}. \tag{2}$$

For all k, it holds that

$$\mathbb{S}_{x_k} \subseteq \mathbb{B}_{\theta_{x_k}}(\bar{\mathbb{P}}_{x_k}),$$

Ambiguity Radii via Local Bounds on Dynamics

Theorem

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For all k, it holds that

$$\mathbb{S}_{x_k} \subseteq \mathbb{B}_{\theta_{x_k}}(\bar{\mathbb{P}}_{x_k}),$$

Proposition

Given $\epsilon > 0$, let $\mathcal{C}_k$ be such that $\theta_{\Delta,k} \leq \epsilon$ for every k. Then,

$$\lim_{k \to \infty} \theta_{x_k} \leq \frac{\theta_\omega}{1 - \mathcal{L}_f} + \frac{\mathcal{L}_f}{1 - \mathcal{L}_f} \epsilon.$$

Evaluating the Method on Bimodal Dynamics

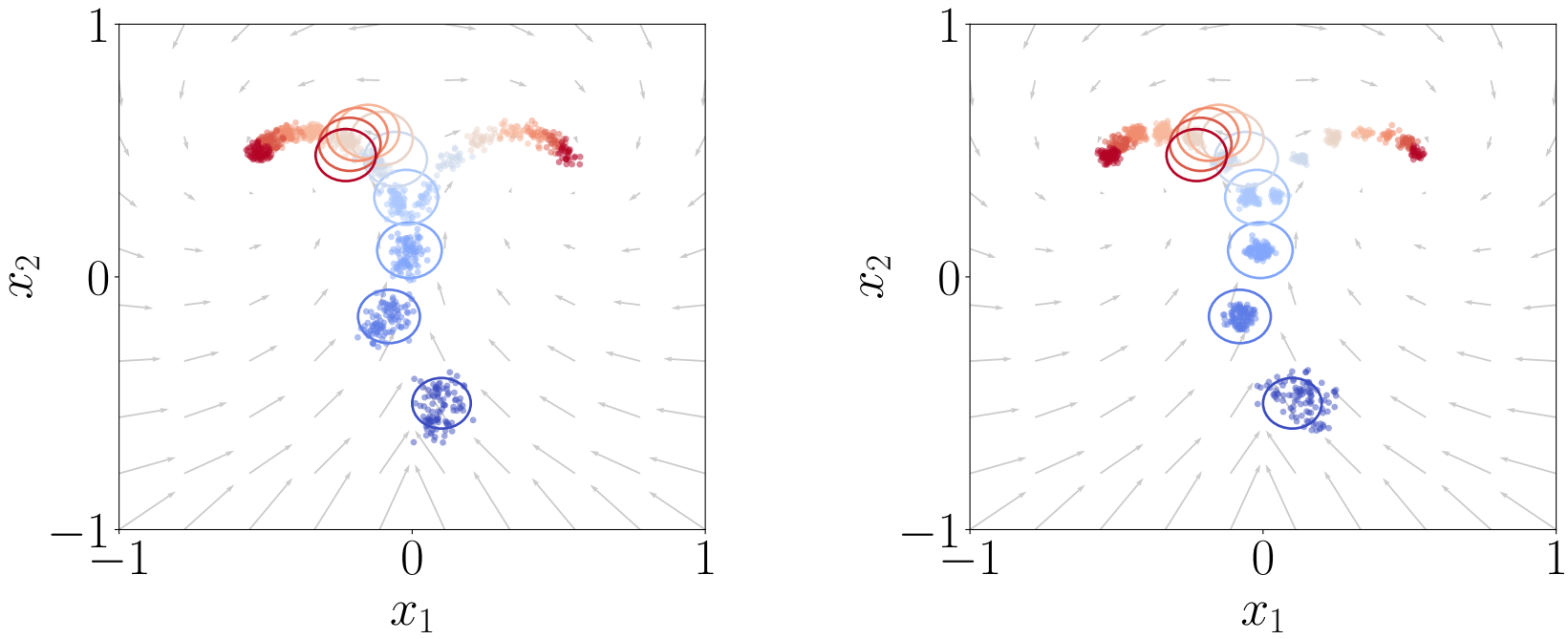


Figure: Samples from $\mathbb{S}_{x_k}$ obtained via MC sampling (**left**), and samples from $\bar{\mathbb{P}}_{x_k}$ (**right**) for $K = 20$ time steps. The circles, centered at the mean of $\bar{\mathbb{P}}_{x_k}$ with radius θ_{x_k}, are guaranteed to contain the mean of the true distributions.

Evaluating on Closed-Loop Switched Dynamics

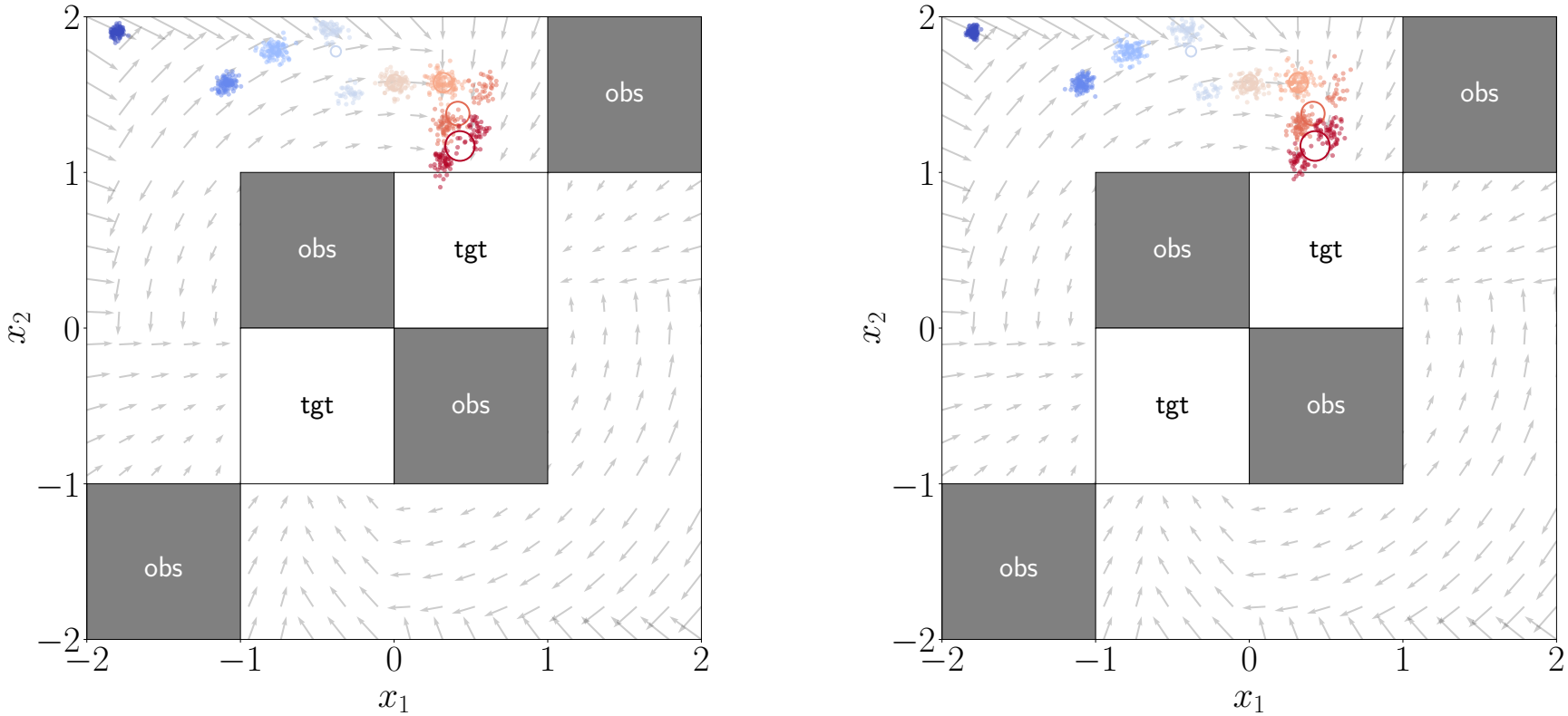


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Summary of Contributions

- A framework for formal uncertainty propagation in non-linear stochastic dynamics with additive noise.
- Handles uncertain initial and noise distributions, covering NN-based and data-driven models.
- Future work: applying the approach within control-design pipelines.

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